

Panel Data Analysis

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2026-09-19

What is panel data?

- **Cross-section:** many units i , each observed **once**.
- **Time series:** **one** unit, many periods t (e.g. French GDP over time).
- **Panel (longitudinal):** the **same** units i followed over periods t — at least **two** dimensions.

Higher dimensions are common:

- exports between i and j , at time t , for product k ;
- wages of workers i in sectors k , cities c , over years t .

Unobserved heterogeneity and omitted-variable bias

We want the causal effect β of x_{it} on y_{it} . In a regression without controls, every unmeasured determinant of y_{it} is in the error:

$$y_{it} = \alpha + \beta x_{it} + \varepsilon_{it}, \quad \varepsilon_{it} = \underbrace{f_i}_{\text{individual}} + \underbrace{f_t}_{\text{time}} + u_{it}.$$

- f_i — **unobserved individual** characteristics: ability, institutions, geography... fixed over time, differing across units.
- f_t — **time** effect: shocks hitting **all** units in period t (crises, world prices, technology).
- If f_i or f_t is correlated with x_{it} , then $\mathbb{E}[\varepsilon_{it} \mid x_{it}] \neq 0$: x_{it} is **endogenous** and OLS is **biased** (omitted-variable bias).

Objective

Observing the *same* units over *several* periods allows us to **remove** f_i and f_t **from the error**, and to identify β even when they are correlated with x .

Correlated errors and inference

The error term has a second problem, on the standard errors (the significance of the coefficients).

- In a panel, two observations can share a component of the error, and their errors are then correlated. Two observations of the **same unit** share f_i , so the errors of a unit may be correlated over time — **autocorrelation**. Two observations of the **same period** share f_t , so the errors of different units may be correlated within a period — **cross-sectional correlation**.
- The observations are then **not independent draws**, and the usual variance formula no longer applies.
- With correlated errors the estimated variance is **biased**, and in the usual case of positive correlation it is **understated**. Standard errors are too small, t too large, and the test **rejects too often**.
- The section *Clustered standard errors* gives the correction.

Balanced vs. unbalanced panels

- **Balanced:** every unit i is observed in **every** period t — exactly $N \times T$ rows (a full rectangle).
- **Unbalanced:** some unit–period cells are **missing** (units enter/exit, attrition) — fewer than $N \times T$ rows.

Why it matters

A couple of results below — notably that the **between** estimator removes the time effect — hold cleanly **only for balanced panels**. Real data are often unbalanced; we indicate where this matters.

Roadmap

- 1 Panel data: dimensions and the model.
- 2 Estimators that **ignore** heterogeneity: pooled OLS and between.
- 3 The **within (fixed-effects)** estimator.
- 4 **Two-way fixed effects**.
- 5 Standard errors: what Ω is (clustering).
- 6 Application: temperature and growth (Dell et al. 2012).

We deliberately skip random effects and the Hausman test. Random effects requires $\text{Cov}(x_{it}, f_i) = 0$ — the very assumption panel data let us **avoid** — so it is rarely defensible in modern applied work.

Section 1

Naive estimators

Pooled OLS

Treat the panel as one big cross-section, leaving **both** unobserved effects in the error:

$$y_{it} = \alpha + \beta x_{it} + \varepsilon_{it}, \quad \varepsilon_{it} = f_i + f_t + u_{it}.$$

- Removes **neither** f_i nor f_t from the error.
- **Consistent only if** $\text{Cov}(x_{it}, f_i) = 0$ **and** $\text{Cov}(x_{it}, f_t) = 0$ — usually too strong.
- ε_{it} shares f_i across all periods, so a unit's errors are **correlated with each other** (autocorrelation), treated in the section *Clustered standard errors*.

The bias

With a single regressor and exogenous idiosyncratic error ($\text{Cov}(x_{it}, u_{it}) = 0$),
$$\text{plim } \hat{\beta}_{\text{pooled}} = \beta + \frac{\text{Cov}(x_{it}, f_i + f_t)}{\text{Var}(x_{it})},$$
 nonzero as soon as x is correlated with *either* unobserved effect. The transformations below remove that covariance.

The between estimator

Run OLS of $\bar{y}_i$ on $\bar{x}_i$ — a *between* regression on the **unit means** (N data points, one per unit) with error $\bar{\varepsilon}_i$:

$$\bar{y}_i = \alpha + \beta \bar{x}_i + \bar{\varepsilon}_i, \quad \hat{\beta}_B = \frac{\sum_i (\bar{x}_i - \bar{x})(\bar{y}_i - \bar{y})}{\sum_i (\bar{x}_i - \bar{x})^2}.$$

To find this error, average the true model over t (f_i is constant, so it stays). In a **balanced panel** the time effects average to the same grand mean for every unit:

$$\bar{\varepsilon}_i = f_i + \bar{f} + \bar{u}_i, \quad \bar{f} = \frac{1}{T} \sum_t f_t.$$

- That $\bar{f}$ is the **same for all units** $\rightarrow$ a constant the intercept absorbs: **between removes f_t** .
- f_i is **still in the error** $\rightarrow$ **biased if $\text{Cov}(\bar{x}_i, f_i) \neq 0$** .
- Uses **only between** variation; a benchmark, rarely the final estimate.
(Unbalanced panel: the average differs across units and does not cancel.)

Section 2

Fixed effects

The within transformation

Start from the model with both effects,

$$y_{it} = \alpha + \beta x_{it} + f_i + f_t + u_{it},$$

and subtract each unit's time-average ($\bar{f} = \frac{1}{T} \sum_t f_t$):

$$y_{it} - \bar{y}_i = \beta (x_{it} - \bar{x}_i) + (f_t - \bar{f}) + (u_{it} - \bar{u}_i).$$

f_i cancels — but f_t does not

f_i is constant over t , so demeaning subtracts it from itself ($f_i - f_i = 0$) and it **disappears**. $\hat{\beta}_{\text{within}}$ is consistent even if $\text{Cov}(x_{it}, f_i) \neq 0$, but the time effect leaves $f_t - \bar{f} \neq 0$.

- OLS on the demeaned data **is** the **within / fixed-effects** estimator,

$$\hat{\beta}_W = \frac{\sum_i \sum_t (x_{it} - \bar{x}_i)(y_{it} - \bar{y}_i)}{\sum_i \sum_t (x_{it} - \bar{x}_i)^2},$$

identified purely from **within-unit variation over time**.

- A common time shock **still biases** it $\Rightarrow$ **TWFE** (next section).

Equivalence: least-squares dummy variables (LSDV)

Demeaning is identical to OLS with **one dummy per unit**: FE treats the unobserved f_i as **parameters to estimate** (one per unit), not as a random draw — each dummy's coefficient is f_i :

$$y_{it} = \beta x_{it} + f_i + u_{it}.$$

i	t	y_{it}	x_{it}	f_{USA}	f_{INDO}	f_{NIGER}
USA	2023	3.0	12	1	0	0
USA	2022	3.2	14	1	0	0
Indonesia	2023	1.0	22	0	1	0
Indonesia	2022	1.3	23	0	1	0
Niger	2022	0.1	28	0	0	1
Niger	2023	0.4	31	0	0	1

Same $\hat{\beta}$ as demeaning (Frisch–Waugh–Lovell). *Within* = looking **inside** each unit at its own evolution over time.

Frisch–Waugh–Lovell: within = partialling out f_i

Demeaning and N dummies give the same $\hat{\beta}$. To see it, collect the unit dummies in a matrix D and define the **residual-maker**

$$M_D = I - D(D'D)^{-1}D'.$$

- M_D is symmetric and idempotent¹ — a **projection**. Applied to any variable it **subtracts that variable's unit means**: M_D is the within transformation.
- **Frisch–Waugh–Lovell**: regressing y on (X, D) gives the **same** $\hat{\beta}$ as regressing the residualized $M_D y$ on $M_D X$:

$$\hat{\beta}_W = (X' M_D X)^{-1} X' M_D y.$$

- So we never invert the huge $N \times N$ dummy block — we demean and run OLS on k columns. This is exactly what `feols/reghdfe` do internally.

¹Idempotent: applying it twice equals applying it once, $M_D M_D = M_D$.

Decomposition of the variance: within and between

Within and between use **two parts of the same variance**. For a balanced panel, the total variation in x decomposes **exactly**:

$$\underbrace{\sum_{it} (x_{it} - \bar{x})^2}_{\text{total}} = \underbrace{\sum_{it} (x_{it} - \bar{x}_i)^2}_{\text{within (over time)}} + \underbrace{T \sum_i (\bar{x}_i - \bar{x})^2}_{\text{between (across units)}}.$$

- **Within** uses only the first piece, **between** only the second.
- Pooled OLS uses **both**: it is the variance-weighted average

$$\hat{\beta}_{\text{pooled}} = w \hat{\beta}_W + (1 - w) \hat{\beta}_B, \quad w = \frac{S_W}{S_W + S_B},$$

with $S_W = \sum_{it} (x_{it} - \bar{x}_i)^2$ and $S_B = T \sum_i (\bar{x}_i - \bar{x})^2$ (the two pieces above — keep the T).

- **Precision**: the within estimator discards the between piece $\Rightarrow$ **larger standard errors** than pooled OLS.

Balanced panel shown; unbalanced replaces T by T_i and weights accordingly.

Assumption and limits of the within estimator

Strict exogeneity (the key assumption)

$\mathbb{E}[u_{it} \mid x_{i1}, \dots, x_{iT}, f_i] = 0$: the shock is uncorrelated with *past, present and future* regressors. (Rules out feedback, e.g. a lagged y on the right-hand side.)

- ✓ Controls for **all** time-invariant confounders, observed or not — they sit in f_i .
- × **Cannot estimate** the effect of any **time-invariant** regressor: if $x_{it} = x_i$ then $x_{it} - \bar{x}_i = 0$ and it drops out (gender, distance, “was colonized”...).
- × Still **vulnerable to time-varying** confounders — those are *not* in f_i .

Section 3

Two-way fixed effects

Two-way fixed effects (TWFE)

The within estimator left $f_t - \bar{f}$ in the equation — a **common shock** each period (crises, world prices, technology) can still bias it. Add a **time effect** f_t beside the unit effect:

$$y_{it} = \alpha + \beta x_{it} + \underbrace{f_i}_{\text{unit}} + \underbrace{f_t}_{\text{time}} + u_{it}.$$

Estimate by **double-demeaning** ($\ddot{x}_{it} = x_{it} - \bar{x}_i - \bar{x}_t + \bar{x}$, same for y) — FWL with both dummy sets:

$$\hat{\beta}_{TWFE} = \frac{\sum_{it} \ddot{x}_{it} \ddot{y}_{it}}{\sum_{it} \ddot{x}_{it}^2}.$$

- Absorbs time-invariant **unit** heterogeneity **and** unit-invariant **time** shocks.
- Stricter control of confounders → the **standard estimator** of applied microeconomics and the basis of difference-in-differences.
- Identified only if $\ddot{x}_{it}$ still **varies** (not collinear with the time dummies); otherwise the denominator is zero.

Summary of the four estimators

The error contains two unobserved effects, f_i and f_t . The four estimators are OLS on transformed data, and the transformation determines **which effect is removed from the error**:

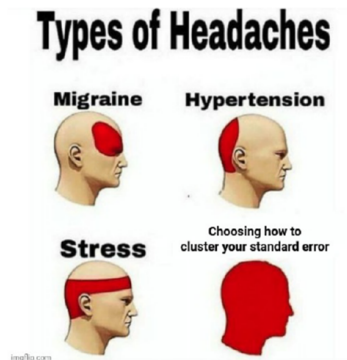
	data fed to OLS	removes f_i ?	removes f_t ?	left biased by
Pooled	raw rows x_{it}	no	no	f_i and f_t
Between	unit means $\bar{x}_i$	no	yes	f_i
Within	$x_{it} - \bar{x}_i$	yes	no	f_t
TWFE	$x_{it} - \bar{x}_i - \bar{x}_t + \bar{x}$	yes	yes	—

- **Between** averages over time $\rightarrow$ removes f_t , keeps f_i .
- **Within** demeanes within units $\rightarrow$ removes f_i , keeps f_t .
- **TWFE** demeanes in both dimensions $\rightarrow$ removes **both**.

*Between removes f_i only in a **balanced** panel; “Within keeps f_t ” means in demeaned form, $f_t - \bar{f}$.*

Section 4

Clustered standard errors



Source: Khoa Vu

What is left in the error after the fixed effects

After within or TWFE, f_i and f_t are no longer in the error, which is u_{it} . What remains to be treated is the correlation of u_{it} between observations. Two pairs matter,

$\text{Cov}(u_{is}, u_{it}) \neq 0$ same unit i , two periods $s \neq t$

$\text{Cov}(u_{it}, u_{jt}) \neq 0$ two units $i \neq j$, same period t

- **Autocorrelation** (first line). A shock to a farm, a firm or a country persists over the following years. $u_{it} = \rho u_{i,t-1} + e_{it}$ with $\rho > 0$ is the usual case.
- **Cross-sectional correlation** (second line). What is left of a common shock after the time effect — a regional drought, a sector price. A shock common to *all* units is already in f_t .
- Clustering by unit allows any correlation of the first kind and assumes the second is zero. The application treats the second (Dell et al.).

If f_i and f_t are kept in the error instead of being removed, they add $\text{Var}(f_i)$ to the first covariance and $\text{Var}(f_t)$ to the second — the random-effects model.

The variance of fixed-effects estimators

The variance of $\hat{\beta}$ depends on the whole covariance matrix of the errors, $\Omega = \mathbb{E}[\varepsilon\varepsilon' \mid X]$. By FWL any FE estimator is OLS on M_D -residualized data, so **X is replaced by $M_D X$** in the usual sandwich formula:

$$\text{Var}(\hat{\beta}_{\text{FE}}) = (X' M_D X)^{-1} (X' M_D \Omega M_D X) (X' M_D X)^{-1},$$

where M_D is the residual-maker defined on the Frisch–Waugh–Lovell slide, built from the dummies of the fixed effects.

- If $\Omega = \sigma^2 I$, this reduces to the classical formula $\sigma^2 (X' M_D X)^{-1}$.
- But with **autocorrelated** errors Ω is **not** scalar and the classical formula does not hold. It **understates** the true variance.
- The correction is to model Ω as block-diagonal by unit — **clustered standard errors**, the rest of this section.

Cluster-robust standard errors

One solution is to cluster the errors. Clustering allows **any** correlation *within* a cluster (a unit across its years) and imposes independence *across* clusters, so Ω is **block-diagonal**, one block Ω_g per cluster:

$$\Omega = \begin{pmatrix} \Omega_1 & & 0 \\ & \ddots & \\ 0 & & \Omega_G \end{pmatrix}$$

The **zeros outside** the blocks assume independence **between clusters**. Correlation is free *inside* a cluster and zero *between* two clusters.

For a unit observed in three periods, its block is 3×3 ,

$$\Omega_1 = \begin{pmatrix} \text{Var}(u_{11}) & \text{Cov}(u_{11}, u_{12}) & \text{Cov}(u_{11}, u_{13}) \\ \text{Cov}(u_{12}, u_{11}) & \text{Var}(u_{12}) & \text{Cov}(u_{12}, u_{13}) \\ \text{Cov}(u_{13}, u_{11}) & \text{Cov}(u_{13}, u_{12}) & \text{Var}(u_{13}) \end{pmatrix}$$

Its diagonal holds the variances, its off-diagonal the **autocorrelations**. The six numbers are left **free** — no AR(1), no equal variances.

The cluster-robust estimator

The numbers inside a block are true variances and covariances, **but** we only have residuals. Liang–Zeger (1986) give an estimator that uses only the residuals of each cluster and is reliable when the clusters are **numerous** (formula in the appendix).

Effective sample size

The precision of a clustered standard error depends on the number of clusters G , not on the NT rows of the data.

Example. 200 people interviewed, 20 in each of 10 villages. If the people of a village share the same local shock, the sample carries about 10 independent observations, not 200.

The Moulton problem (Moulton, 1990)

The problem arises when the regressor is defined at a **coarser level** than the outcome. With units i observed at t and a regressor defined for the region or sector $g(i)$ that contains unit i ,

$$y_{it} = \alpha + \beta x_{g(i)t} + \varepsilon_{it},$$

x takes the same value for every unit of the region or sector (a regional tax, a sector price), while y_{it} varies unit by unit.

- The errors of the units of one region or sector share its shock. With clusters of size $\bar{n}$ and an intraclass correlation ρ of the errors, the true variance of $\hat{\beta}$ exceeds the one OLS prints by the **Moulton factor**

$$\frac{\text{Var}(\hat{\beta})_{\text{true}}}{\text{Var}(\hat{\beta})_{\text{naive}}} = 1 + (\bar{n} - 1)\rho.$$

- The factor grows with the **size of the clusters**. Adding units inside a region or sector raises it instead of improving precision. Cluster at the level where the regressor varies.

Level and number of clusters

- **Cluster at the level of treatment assignment** (or wherever residuals stay correlated). With repeated observations per unit, cluster **by the unit**, not unit–year — the errors are serially correlated over time (Bertrand et al. 2004).
- **Coarser is safer** (weakly larger SE) — but you need **enough clusters**.
- **Too few clusters** $\Rightarrow$ the formula is unreliable: a **wild-cluster bootstrap** (simulate the null instead of using the t -table) restores the size of the test — see the appendix.
- Default: with grouped/repeated data, **cluster by the group** — and report G .

Section 5

Application

Temperature and economic growth

Does a hot climate lower economic growth?

There are countries where the excess heat ... makes men lazy and discouraged.

— Montesquieu (1748)

The modern hypothesis: high temperatures reduce productivity and growth.
The question is how to test it with a **credible identification** (Dell et al. 2012).

Cross-section: time-invariant confounders

A cross-section (or between) regression of growth on temperature,

$$\text{Growth}_i = \gamma \text{Temp}_i + \varepsilon_i,$$

compares **hot vs. cold countries**. This is a problem because countries differ in many *fixed* ways (f_i):

- hot, poor countries were often **colonized**, with lasting institutional damage;
- the industrial revolutions happened in **temperate** climates (accident + slow diffusion).

These are **time-invariant confounders** → the cross-section / between estimate is biased.

Within: temperature variation inside each country

Add a country fixed effect and use **year-to-year temperature variation within each country**:

$$\text{Growth}_{it} = \beta \text{Temp}_{it} + f_i + u_{it}.$$

- f_i absorbs colonization, institutions, geography — *every* fixed country trait.
- Identification: does a country grow **less in its own hotter years**? (data: many countries, 1950–2003.)
- Still a threat: **time-varying** confounders correlated with temperature (El Niño cycles, adaptation) — a country fixed effect cannot remove those.

Results: average effect

Annual growth rate	
Temperature	−0.325 (0.285)

Country fixed effects; SE clustered by country.

- Negative but **statistically insignificant**: temperature fluctuations do not move growth *on average across all countries*.
- But an average can hide **heterogeneity**...

Heterogeneity: poor vs. rich

- Poor countries are concentrated in **agriculture** (up to ~80% of activity), poorly insured.
- Temperature should have a **larger** effect there than in rich, service-based, air-conditioned economies.
- Test it with an **interaction**: let the temperature effect differ for poor countries.
- We also add **year effects** f_t — global shocks (recessions, oil prices) common to all countries would otherwise bias the estimate.

Interaction: the effect in poor countries

With both country and year fixed effects (f_i, f_t):

$$g_{it} = f_i + f_t + \beta T_{it} + \alpha (T_{it} \times POOR_i) + u_{it}.$$

The marginal effect of temperature **depends on** $POOR_i$ — so you **add** the two coefficients for poor countries:

$$\frac{\partial g_{it}}{\partial T_{it}} = \beta + \alpha POOR_i = \begin{cases} \beta & \text{rich } (POOR = 0) \\ \beta + \alpha & \text{poor } (POOR = 1). \end{cases}$$

	Annual growth rate
Temperature	0.261 (0.312)
Temperature $\times$ Poor	−1.655*** (0.485)

So the poor-country effect is $\beta + \alpha = 0.261 - 1.655 = -1.394$ pp — vs ≈ 0 in rich countries.

Country & year FE; clustered by country. Testing $\beta + \alpha$ needs its **own** SE (a linear-combination test), not the stars on α or on β .

Beyond Dell et al.: inference in country-year panels

Dell et al. cluster by country, then two-way by country and region-year. Clustering by country treats the correlation of one country's errors over time. What it leaves untreated is the correlation between **neighbouring countries in the same year**, which a drought or an El Niño year produces. The estimators below treat that second direction.

- **Serial correlation inside a country: cluster by country.** 125 clusters is enough.
- **Spatial correlation between countries: Conley (1999) standard errors.** The correlation of shocks decays with distance, not at a region border. Each pair of observations enters Ω with a weight that declines with the distance between the units, up to a cutoff (1 000 km, say). A distance cutoff replaces the choice of a cluster. Standard in the climate–economy literature since Hsiang (2016).
- **Long panels: Driscoll–Kraay (1998) standard errors** allow any cross-sectional correlation but need a long T ($T \approx 50$ here). A cross-check of Conley.
- **Two-way clustering by country and region-year** has few groups in the second dimension; its variance can fail to be positive definite (MacKinnon et al. 2021). Prefer Conley when coordinates exist.
- **Whatever the choice, check the size.** Reassign the weather series across countries at random, re-estimate, count the rejections at 5 %: Dell et al. found 4 %.

Section 6

Python

The toolkit and long format

Panel methods here use `pyfixest` — it *absorbs* fixed effects and clusters in one call (like Stata's `reghdfe` or R's `fixest`). Data must be in **long format**: one row per unit–period, with an `id` and a `time` column.

country	year	growth	temp
USA	2000	3.0	12
USA	2001	2.6	13
Niger	2000	0.5	28
Niger	2001	0.2	30

(Wide data — one column per year — must be **reshaped** to long first.)

Load the panel and check its structure

Four lines open the file and describe how many periods each unit is observed.

```
import pandas as pd, pyfixest as pf
df = pd.read_csv("rica_panel.csv")      # 1
n_t = df.groupby("id").size()           # 2
df["id"].nunique()                       # 3
n_t.describe()                          # 4
```

- 1 — reads the file into a table `df` in long format: one row per farm and year.
- 2 — counts the rows of each farm, that is T_i . `groupby("id")` splits the table by farm, `.size()` counts the rows of each group; `n_t` has one value per farm.
- 3 — the number of distinct farms, N .
- 4 — the distribution of T_i : a minimum of 1 signals **singletons**, farms seen once, which carry no within variation and will be dropped by the fixed effects.

Within and between variation of a variable

Five lines split the variation of x into its within and between parts.

```
moy_i = df.groupby("id")["x"].transform("mean")      # 1
within = df["x"] - moy_i                             # 2
S_W = (within ** 2).sum()                             # 3
S_B = ((moy_i - df["x"].mean()) ** 2).sum()           # 4
w = S_W / (S_W + S_B)                                # 5
```

- 1 — the mean of x in each farm, written on **every row of that farm**; this is what `.transform("mean")` does, unlike `.mean()`, which would give one row per farm.
- 2 — each observation minus its farm's mean: the within deviation.
- 3 — the within sum of squares S_W .
- 4 — the between sum of squares S_B : each farm's mean minus the overall mean, squared and summed, so a farm counts T_i times.
- 5 — the within share w of the chapter. A small w means the estimator that uses within variation only works with a small part of the data.

The four estimators with one function

Seven lines estimate pooled, between, within and TWFE, with standard errors clustered by farm (line 1, stored once), and put them side by side.

```
cl = {"CRV1": "id"} # 1
m_pool = pf.feols("y ~ x + C(region)", df, vcov=cl) # 2
moy = df.groupby("id")[["y", "x"]].mean() # 3
m_betw = pf.feols("y ~ x", moy, vcov="hetero") # 4
m_within = pf.feols("y ~ x | id", df, vcov=cl) # 5
m_twfe = pf.feols("y ~ x | id + annee", df, vcov=cl) # 6
pf.etable([m_pool, m_betw, m_within, m_twfe]) # 7
```

- 2 — pooled OLS: all rows together, region dummies as controls.
- 3 and 4 — between: moy has one row per farm, the mean of y and x; the regression on these means uses cross-farm variation only.
- 5 — within: after |, id names the fixed effect to absorb, f_i .
- 6 — TWFE: id + annee absorbs f_i and f_t ; singleton farms are dropped.
- 7 — the four results in one table, to watch $\hat{\beta}$ change as each effect enters.

Read the results and the fixed effects

Six lines pull out what the chapter asks for: coefficients, the year effects, residuals.

```
b, se = m_twfe.coef()["x"], m_twfe.se()["x"]      # 1
ci = m_twfe.confint().loc["x"]                  # 2
t = (b - 0.136) / se                             # 3
fe = pd.Series(m_twfe.fixef()["C(annee)"])       # 4
fe.loc[2000] = 0.0                               # 5
u = m_twfe.resid()                              # 6
```

- 1 and 2 — the coefficient, its standard error and its 95 % interval. In `pyfixest` these are **methods**, with parentheses, unlike `statsmodels` attributes.
- 3 — the t of $H_0 : \beta = 0,136$, a cost share; the summary tests zero only.
- 4 — the estimated year effects $\hat{f}_t$, a dictionary keyed by the fixed-effect name, turned into a series indexed by year.
- 5 — the reference year is absorbed and not returned; adding it as zero lets you read every $\hat{f}_t$ as a difference from 2000.
- 6 — the residuals, one per observation the estimation used, in the same order.

Four standard errors for one coefficient

Three lines re-estimate the same model under four assumptions on the error covariance.

```
for v in ["iid", "hetero", {"CRV1": "id"}, {"CRV1": "region"}]: # 1
    m = pf.feols("y ~ x | id + annee", df, vcov=v) # 2
    print(m.se()["x"], m._G) # 3
```

- 1 — the four choices of Ω from the appendix: classical, heteroskedastic, clustered by farm, clustered by region.
- 2 — `vcov` is the only thing that changes; $\hat{\beta}$ is identical each time.
- 3 — the standard error under each choice, and `_G`, the number of clusters the estimation actually used, after dropping singletons. The Moulton factor of the chapter is checked against the ratio of the third to the second.

Serial correlation of a farm's residuals

Four lines measure the year-to-year correlation of a farm's residuals.

```
r = df_used[["id", "annee"]].copy() # 1
r["u"] = m_twfe.resid() # 2
r["u_lag"] = r.groupby("id")["u"].shift(1) # 3
rho = r[["u", "u_lag"]].corr().iloc[0, 1] # 4
```

- 1 — a small table with the farm and the year of every observation the estimation used; `df_used` is `df` without the singletons, so that it lines up with the residuals.
- 2 — attaches the residuals to it.
- 3 — the residual of the **previous row of the same farm**: `groupby("id")` keeps the shift inside each farm, `shift(1)` moves values down one row. Sort by farm and year first, and keep only consecutive years.
- 4 — the correlation between a residual and its lag, ρ . Positive means a farm's observations are not independent draws, which is what clustering accounts for.

References I

For a fuller textbook treatment of every method in this chapter, see Wooldridge (2025).

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Section 7

Appendix

The covariance Ω : what we assume

Everything is the **sandwich** $\text{Var}(\hat{\beta}) = (X'X)^{-1}X'\Omega X(X'X)^{-1}$, with $\Omega = \mathbb{E}[\varepsilon\varepsilon' \mid X]$ the $NT \times NT$ error covariance. Only the **meat** $X'\Omega X$ changes with our assumption:

Three choices for Ω — same $\hat{\beta}$, different SE

i.i.d.: $\Omega = \sigma^2 I$ (classical). **Heteroskedastic:** $\Omega = \text{diag}(\sigma_i^2)$ (White/HC).
Clustered: Ω block-diagonal (free within groups).

The point estimate is the same under the three choices; only the standard error depends on Ω .

The cluster-robust estimator (Liang–Zeger, 1986)

With G clusters — errors free *within*, independent *across*:

$$X' \Omega X = \sum_g X'_g \Omega_g X_g \approx \sum_g X'_g \hat{\varepsilon}_g \hat{\varepsilon}'_g X_g,$$

$$\widehat{\text{Var}}_{\text{CR}}(\hat{\beta}) = \frac{G}{G-1} (X'X)^{-1} \left(\sum_g X'_g \hat{\varepsilon}_g \hat{\varepsilon}'_g X_g \right) (X'X)^{-1}.$$

- Each cluster's residual outer product is a *poor* estimate of Ω_g — but the **sum over independent clusters** is consistent. $\frac{G}{G-1}$ is a small-sample fix.
- Consistent as $G \rightarrow \infty$ (**many clusters**), *not* as the cluster size grows.

Bertrand et al. (2004): the experiment

A simulation test of inference in policy evaluations. The question: do the **conventional standard errors** used in these studies give tests of the right size?

- **Data.** The *Current Population Survey* (CPS) — the large monthly U.S. household survey — gives many years of individual wages across the 50 states.
- **Placebo law.** They invent a **fake** state policy: pick a random state and a random year, and pretend a law switched on there. It changes nothing, so the true effect is **zero by construction** ($\beta = 0$).
- **The test.** Estimate the “effect” of this fake law on wages and ask how often the t -test rejects $\beta = 0$ at the 5% level. With correct SEs it should reject $\approx 5\%$ of the time. Repeat over hundreds of random placebos.

Bertrand et al. (2004): results and correction

The result

Classical (i.i.d.) SEs rejected the true null in $\approx 45\%$ of placebos — nine times the nominal 5%. Clustering by **state** brought it back to 6–8%.

- **Reason.** Wages are **serially correlated** within a state (this year looks like last year), *and* the fake law, once on, **stays on** for years (treatment is persistent). The many state–year rows are far from independent — effectively there is only **one experiment per state**, not one per state–year.
- **Correction.** Cluster at the **state** level (the unit of treatment): the SEs then count ≈ 50 blocks, not thousands of rows, and the test size is restored.
- **Lesson.** Cluster at the level of treatment assignment — not the finer unit–year.

Abadie et al. (2023): clustering as a design question

Clustering is decided by the **design** of the data. Cluster if **either**:

- ➊ **Sampling design** — you sampled clusters, then units within them (schools, then pupils).
 - ➋ **Assignment design** — treatment is assigned at the cluster level.
- If **neither** holds (a census; unit-specific treatment), clustering may be unnecessary and can even *cost* precision.
 - The criterion is whether the sample or the treatment was drawn by cluster; a correlation of the residuals alone does not require clustering.

Where to cluster: the practical rule

Rule

Cluster at the **coarsest** level at which (i) treatment is assigned / sampling was done, or (ii) residuals stay correlated.

- Nested clusterings $C_1 \subset C_2$: $\widehat{\text{Var}}_{C_2} \geq \widehat{\text{Var}}_{C_1}$ — **coarser is (weakly) larger**, hence safer for test size.
- But coarser shrinks G ; below ~ 15 – 20 clusters the formula degrades.
- **Never cluster finer than the level of treatment assignment.**

Multiway clustering (Cameron et al., 2011)

Correlation along **two** non-nested dimensions (e.g. country *and* year):

$$\widehat{\text{Var}}_{2\text{-way}} = \widehat{\text{Var}}_{\text{country}} + \widehat{\text{Var}}_{\text{year}} - \widehat{\text{Var}}_{\text{country} \cap \text{year}}.$$

- Each piece is a valid one-way CRV; subtracting the intersection avoids double-counting.
- Use only with a clear reason for two dimensions; **unstable** when one dimension has few clusters (e.g. few years).

Few clusters: two distinct problems

CRV1 is consistent only as $G \rightarrow \infty$. When G is small it misbehaves in **two** ways:

- ➊ **Reference distribution.** Compare t to t_{G-1} , not the normal (e.g. $t_{21,0.975} = 2.08$ vs 1.96) — software does this automatically.
- ➋ **Noisy variance.** $\sum_g X'_g \hat{\varepsilon}_g \hat{\varepsilon}'_g X_g$ is itself a noisy estimate and **under-covers**, worse with unbalanced clusters.

Each needs a *different* remedy.

Few clusters: corrections

Rather than trust the t -table when G is small, **simulate** the test's own distribution and read the p -value off it.

- **Wild-cluster bootstrap** (Cameron et al. 2008): re-estimate the model thousands of times, each time flipping the sign of every cluster's residuals at random (± 1). This builds the distribution of the t -statistic *under the null*, using only G blocks. The former default; it **fails with few treated clusters** (p -values close to 1).
- **CR2 + Satterthwaite df** (Imbens–Kolesár 2016; Pustejovsky–Tipton 2018): bias-corrected sandwich + effective df. Recommended down to $G \approx 10$ when leverage is balanced.
- **CR3 cluster jackknife** (MacKinnon et al. 2023): robust when a few clusters carry most of the leverage.
- **Randomization inference** (Young 2019): permute treatment across clusters; distribution-free.

Summary, and further reading

Decision tree

Assigned/sampled in clusters, or residuals correlated? → cluster there. Coarser = safer, but keep enough clusters. Few clusters → CR2 / CR3 or randomization inference, **not** a naive wild bootstrap when treatment is concentrated.

Moulton (1990); Liang–Zeger (1986); Bertrand et al. (2004); Cameron et al. (2008, 2011); Cameron–Miller (2015); Imbens–Kolesár (2016); Pustejovsky–Tipton (2018); Abadie et al. (2023); MacKinnon et al. (2023); Young (2019).